Two Paths, One Evolution: Testing the Extended Phenotype Theory Across Legal Systems

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Date: August, 2025
Version: 1.0

Abstract

This article extends the extended phenotype theory of legal evolution to comparative law, proposing that common law and civil law systems represent distinct evolutionary mechanisms responding to similar selective pressures. Drawing on Dawkins' concept of extended phenotypes and recent developments in legal memetics, I argue that the persistent convergence between these supposedly incompatible legal traditions reflects not borrowing or globalization but parallel evolution—different paths to similar adaptive solutions.

Common law exhibits continuous micro-evolution through judicial precedent, while civil law follows punctuated equilibrium with long stasis periods interrupted by comprehensive codification. Both mechanisms, however, channel pre-existing legal concepts (memes) whose survival depends on environmental fitness rather than conscious design. Louisiana's successful hybridization over two centuries demonstrates evolved "universal receptor capacity," contrasting sharply with Argentina's chronic rejection of incompatible transplants attempted without evolutionary adaptation.

The theory generates testable predictions: (1) common law systems should adapt to crises faster than civil law systems; (2) convergence should be strongest in globally competitive fields; (3) transplant success should correlate with evolutionary distance between legal systems; (4) simple, portable legal concepts should spread more successfully than complex, culturally embedded ones. Preliminary evidence from COVID-19 responses and financial crisis adaptations supports these predictions.

If legal evolution operates universally through varying mechanisms, the implications for comparative law are profound. Legal transplants require not quality assessment but compatibility analysis. Convergence reflects not cultural hegemony but adaptive responses to universal coordination problems. The common law/civil law

distinction becomes less fundamental than traditionally assumed—merely alternative evolutionary strategies, like sexual versus asexual reproduction in biology.

Keywords: legal evolution, extended phenotype, common law, civil law, legal transplants, comparative law, memetics, punctuated equilibrium, convergent evolution, Louisiana

JEL Classification: K00, K10, K33, K40, B15, B52, P48

"I am inclined to believe that in nearly the same way as two men have sometimes independently hit on the very same invention, so natural selection... has sometimes acted on the same lines in different organic beings."

— Charles Darwin, On the Origin of Species (1859)

I. Introduction

Darwin's observation about independent invention captures a persistent puzzle in comparative law. Common law and civil law systems developed through fundamentally different methods—one through judicial precedent, the other through comprehensive codes—yet they repeatedly converge on remarkably similar solutions. Contract law, tort liability, corporate structures, bankruptcy priorities: the substantive outcomes align far more than their divergent methodologies would predict.

This convergence has been documented but rarely explained. Traditional accounts invoke borrowing, colonialism, or globalization. These factors undoubtedly matter, but they fail to explain why certain transplants succeed while others fail catastrophically, or why convergence occurs even in isolated jurisdictions with minimal external contact.

This article proposes that common law and civil law represent two distinct evolutionary mechanisms responding to similar selective pressures. If the theory of legislators as extended phenotypes developed in my recent analysis of Argentine legal evolution (Lerer, 2025) holds true, it cannot be unique to any single legal tradition. Evolution, if it operates in law at all, must operate universally—though through varying mechanisms.

The extended phenotype framework suggests that legal actors—legislators, judges, administrators—do not consciously design law but rather serve as vehicles through which pre-existing legal concepts (memes) replicate and spread based on their fitness in particular environments. For operational purposes, a legal meme is defined as any discrete legal concept that can be: (1) identified in multiple jurisdictions, (2) traced through time, (3) measured for prevalence. Examples include "good faith" in contracts, "reasonable person" standard, or "force majeure" clauses. Memetic fitness is measured by: (a) adoption rate across jurisdictions, (b) persistence over time, (c) resistance to replacement. A meme with high fitness appears in more than 50% of surveyed jurisdictions and persists more than 50 years without substantial modification.

II. The Evolutionary Hypothesis

The core distinction between common law and civil law may not be evolution versus design, as traditionally conceived, but rather two different modes of evolution: continuous variation with selection versus punctuated equilibrium.

Common law operates through constant micro-evolution. Each case introduces potential variations through lawyers' arguments. Judges select among these variations, establishing precedents that become heritable through stare decisis. Subsequent cases test these precedents, occasionally overruling them when environmental pressures shift. The process is visibly Darwinian: variation, selection, inheritance, modification.

Civil law follows a pattern evolutionary biologists call punctuated equilibrium. Long periods of stasis, where the code remains unchanged despite accumulating tensions, punctuate with sudden comprehensive reforms. The Napoleonic Code governed for a century before replacement. The Argentine Civil Code lasted 146 years before the 2015 reform changed sixty percent of its content virtually overnight. Between these dramatic speciation events, the system remains remarkably stable, with change occurring through interpretive drift rather than formal amendment.

Neither system involves conscious design in the way their architects imagine. As demonstrated in the Argentine case (Lerer, 2025), legislators cannot anticipate how their codes will interact with unforeseen circumstances any more than common law judges can predict the long-term consequences of their precedents. Both systems evolve through trial and error, success and failure, survival and extinction of legal concepts. The mechanisms differ; the underlying process remains evolutionary.

The memetic framework developed in my previous work applies to both systems. Legal concepts—from "reasonable person" in common law to "buena fe" in civil law—function as memes that replicate through legal education, judicial decisions, and professional practice. Their survival depends not on their truth or justice but on their fitness: their ability to solve coordination problems, satisfy powerful interests, and adapt to changing circumstances.

III. Louisiana: The Natural Experiment

Louisiana presents what Julio Cueto Rúa identified decades ago as a unique laboratory for understanding legal compatibility. A civil law system embedded within a common law nation, Louisiana neither adopted nor rejected American legal concepts—it metabolized them into something distinctly its own.

Consider the doctrine of abuse of rights, central to civil law but alien to common law. When similar cases arose in Louisiana, the courts reached functionally identical outcomes without ever invoking the continental doctrine. In Higgins Oil & Fuel Co. v. Guaranty Oil Co. (1919), the Louisiana Supreme Court prevented a landowner from maintaining an unused well that damaged his neighbor's oil extraction. The court didn't need to invoke "abus de droit"—it found other doctrinal paths to the same destination.

This is not borrowing or resistance but something more fundamental: parallel evolution. Louisiana developed, over two centuries, what might be termed universal receptor capacity—the ability to process legal concepts from either tradition without immunological rejection. The system evolved this capacity not through conscious design but through generations of

lawyers and judges trained in both traditions, creating doctrinal bridges that didn't exist in either parent system.

Cueto Rúa documented this phenomenon without the evolutionary framework to explain it. He saw Louisiana achieving what Argentina could not: successful integration of incompatible traditions. The difference wasn't planning or wisdom but evolutionary time. Louisiana had two hundred years of gradual adaptation. Argentina attempted the same integration overnight.

Beyond Louisiana and Argentina, preliminary observation suggests similar patterns in other hybrid systems. Quebec maintains French civil law within Canadian common law. Scotland preserves Roman-Dutch elements within the UK. South Africa combines Roman-Dutch with English law. Japan integrated German civil law while maintaining indigenous elements. Each represents varying degrees of evolutionary adaptation, from Quebec's successful compartmentalization to Japan's selective absorption. Systematic study of these cases remains for future research.

This aligns perfectly with the extended phenotype theory. Louisiana's legal actors became vehicles for both common law and civil law memes, developing the capacity to express either depending on context. The system didn't choose between traditions; it evolved to metabolize both.

IV. Preliminary Evidence and Testable Predictions

The evidence for legal evolution remains preliminary, drawn from a limited sample of jurisdictions. Yet certain patterns appear consistently enough to warrant hypothesis formation.

The response to the COVID-19 pandemic provides a natural experiment. Courts worldwide faced unprecedented contract disputes as lockdowns made performance impossible. Without coordination, jurisdictions converged on remarkably similar solutions. Common law courts stretched force majeure and frustration doctrines. Civil law systems applied imprevisión and

rebus sic stantibus. The terminology differed; the outcomes aligned. German courts reached similar conclusions to New York courts, which paralleled decisions in São Paulo and Tokyo.

This convergence suggests that similar environmental pressures—a global pandemic disrupting contracts—selected for similar legal adaptations regardless of the underlying system. The specific doctrinal path mattered less than the functional outcome: some mechanism to adjust or excuse contractual obligations under extraordinary circumstances.

From these observations and the theoretical framework established in my analysis of legal memetics (Lerer, 2025), several predictions follow:

First, common law systems should respond to crises faster than civil law systems, as their continuous evolution allows immediate adaptation while civil law must await legislative intervention. The 2008 financial crisis offers a test: common law jurisdictions modified debtor-creditor relations through case law within months, while civil law countries required years to enact statutory reforms.

Second, convergence should be strongest in areas subject to global competitive pressures—commercial law, intellectual property, financial regulation—and weakest in culturally embedded fields like family law or inheritance. A systematic comparison of similarity indices across legal domains could test this prediction.

Third, successful legal transplants should correlate with what might be termed "evolutionary distance" between donor and recipient systems. Transplants between similar systems (German to Austrian) should succeed more frequently than between distant systems (American to Japanese), controlling for other factors like economic development and colonial history.

Fourth, building on the concept of "memetic fitness" developed in my previous work, legal concepts with simple, portable structures (like basic contract principles) should spread more successfully than complex, culturally embedded ones (like trust law), regardless of the transplant direction.

V. The Question of Generalizability

Tal Yarkoni recently identified what he calls the "generalizability crisis" in psychology: researchers make sweeping theoretical claims based on narrow empirical foundations (Yarkoni, 2022). The same challenge confronts this analysis. I observe patterns in Argentina, Louisiana, and a handful of other jurisdictions, then propose universal principles of legal evolution.

This expansive theorizing from limited data is deliberate and necessary. Darwin published Origin of Species despite vast gaps in the fossil record. Wallace reached identical conclusions from different evidence. Sometimes theoretical boldness must precede empirical completeness, particularly when the theory generates testable predictions that can guide future research.

The claim is not that legal evolution is proven but that it offers a coherent explanation for otherwise puzzling phenomena: spontaneous convergence, successful and failed transplants, the persistence of obviously dysfunctional rules alongside rapid change in other areas. If the theory is wrong, testing will reveal its failures. If it's right, we gain a powerful framework for understanding legal change.

Daniel Kahneman warned of "theory-induced blindness"—once we adopt a theoretical framework, it becomes difficult to see its flaws (Kahneman, 2011, p. 277). Having spent years developing the extended phenotype theory of law (Lerer, 2025), I may now see evolution where others see conscious design, memetic replication where others see rational choice. This is a risk inherent in theoretical work. The response is not to abandon theory but to make it testable.

VI. Implications for Comparative Law

Should the evolutionary hypothesis prove valid, several implications follow for comparative legal scholarship and practice.

The traditional distinction between common law and civil law becomes less fundamental than typically assumed. Rather than essentially different systems, they represent alternative evolutionary strategies—sexual versus asexual reproduction in biology offers an apt parallel. Both can produce complex, adaptive organisms; they simply generate variation through different mechanisms.

Legal transplants require reconsideration through an evolutionary lens. As demonstrated in the Argentine case study (Lerer, 2025), success depends not on the quality of the transplanted rule but on compatibility between legal environments. A perfectly functional German environmental regulation may fail in Argentina not because Argentina is deficient but because the regulatory organism evolved for a different ecosystem. The solution isn't better transplants but evolved compatibility—either adapting the transplant to local conditions or developing the local capacity to metabolize foreign legal concepts.

The convergence phenomenon suggests that certain legal solutions possess what biologists call convergent evolution—independent development of similar features in response to similar environmental pressures. Eyes evolved independently dozens of times because sight provides enormous adaptive advantage. Perhaps fundamental legal concepts like property rights, contract enforcement, and dispute resolution mechanisms converge because they solve universal problems of human coordination.

Héctor Mairal identified decades ago what he called a "curious paradox" in Argentine law: a constitution inspired by the United States interpreted through French administrative law (Mairal, 1984, vol. I, p. XVIII). Manuel García Mansilla described this as forcing a square peg into a round hole (García Mansilla, 2014, p. 650). Through the evolutionary lens, this isn't paradox but predictable incompatibility—an attempted transplant between systems too evolutionarily distant for successful integration without extended adaptation.

VII. A Research Agenda

This article proposes a hypothesis, not a conclusion. Testing it requires systematic investigation across multiple jurisdictions, tracking legal change over extended periods, and developing metrics for legal similarity and divergence.

The immediate research priorities include:

Documentation of convergence patterns across fifty or more jurisdictions, controlling for direct borrowing and colonial influence. This requires collaboration among comparativists with deep knowledge of local systems.

Development of quantitative measures for legal similarity that go beyond surface features to capture functional equivalence. Natural language processing and machine learning offer promising methodologies.

Historical analysis of legal change to identify evolutionary patterns—gradual adaptation versus punctuated change, successful versus failed variations, extinction and revival of legal concepts.

Experimental studies of legal transplants, tracking success and failure rates while measuring relevant variables like institutional capacity, cultural distance, and economic pressures.

Application of the memetic framework from my previous work to specific doctrinal areas, testing whether legal concepts that survive and spread share identifiable characteristics that distinguish them from those that fail or remain localized.

VIII. Conclusion

The extended phenotype theory, as developed in my recent analysis of Argentine legal evolution (Lerer, 2025), suggests that legislators and judges don't design law so much as channel pre-existing legal concepts whose survival depends on environmental fitness. If valid, this framework should apply across legal traditions, though through different mechanisms.

Common law and civil law may represent parallel experiments in legal evolution—one achieving adaptation through continuous judicial selection, the other through periodic legislative mutation. Their convergence despite methodological divergence suggests that legal evolution, like biological evolution, finds multiple paths to similar solutions when facing comparable environmental pressures.

Louisiana demonstrates that hybrid systems can evolve receptor capacity for both traditions, though the process requires generations rather than years. Argentina illustrates the cost of attempting instant compatibility—chronic rejection that persists decades after the initial transplant, as documented in my analysis of the 1981 mortgage crisis and its legal aftermath (Lerer, 2025).

These observations generate testable predictions about the speed of legal change, patterns of convergence, and transplant success rates. Whether these predictions hold across broader samples remains to be determined. The theory could be wrong in general or right only in specific contexts. Only systematic testing will tell.

What seems clear is that legal change exhibits patterns that transcend conscious design. Whether we call it evolution or something else, understanding these patterns requires moving beyond traditional doctrinal analysis to examine how legal systems adapt, survive, and occasionally transform in response to changing environments. Darwin and Wallace independently discovered natural selection because the evidence, properly considered, pointed toward evolution. Perhaps legal scholars are approaching a similar convergence—not through coordination but because the patterns in legal development, properly considered, point toward evolutionary processes we're only beginning to understand.

The extended phenotype framework offers one way to understand these processes. Legal actors—whether common law judges or civil law legislators—may be less authors of law than vehicles through which legal memes compete, replicate, and evolve. If so, the distinction between common law and civil law becomes less about conscious methodology and more about the mechanism through which the same fundamental evolutionary process operates.

This is not legal nihilism or determinism. Understanding evolution doesn't eliminate agency; it clarifies where agency can be effective. A gardener who understands plant evolution gardens better than one who doesn't. Perhaps legal systems that understand their own evolutionary nature can reform more effectively than those that persist in the illusion of pure rational design.

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Author Note: This article builds on the theoretical framework developed in "The Legislator as Extended Phenotype" (Lerer, 2025), extending its analysis from crisis responses in a single civil law jurisdiction to comparative examination of evolutionary mechanisms across legal traditions.

Acknowledgments: This article builds on my previous work "The Legislator as Extended Phenotype: A Darwinian Theory of Legal Evolution." All errors remain my own.

Declaration: No funding was received for this research. The author declares no conflicts of interest.

Version History: Version 1.0 - January 2025

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Lerer, Ignacio Adrian, Two Paths, One Evolution: Testing the Extended Phenotype Theory Across Legal Systems (August 13, 2025). Available at

SSRN: https://ssrn.com/abstract=5391036 or http://dx.doi.org/10.2139/ssrn.5391036